

Digital Forensics and AI-Based Cybercrime Investigation

CSA1003 – Software Engineering

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Introduction to the Emerging Technology



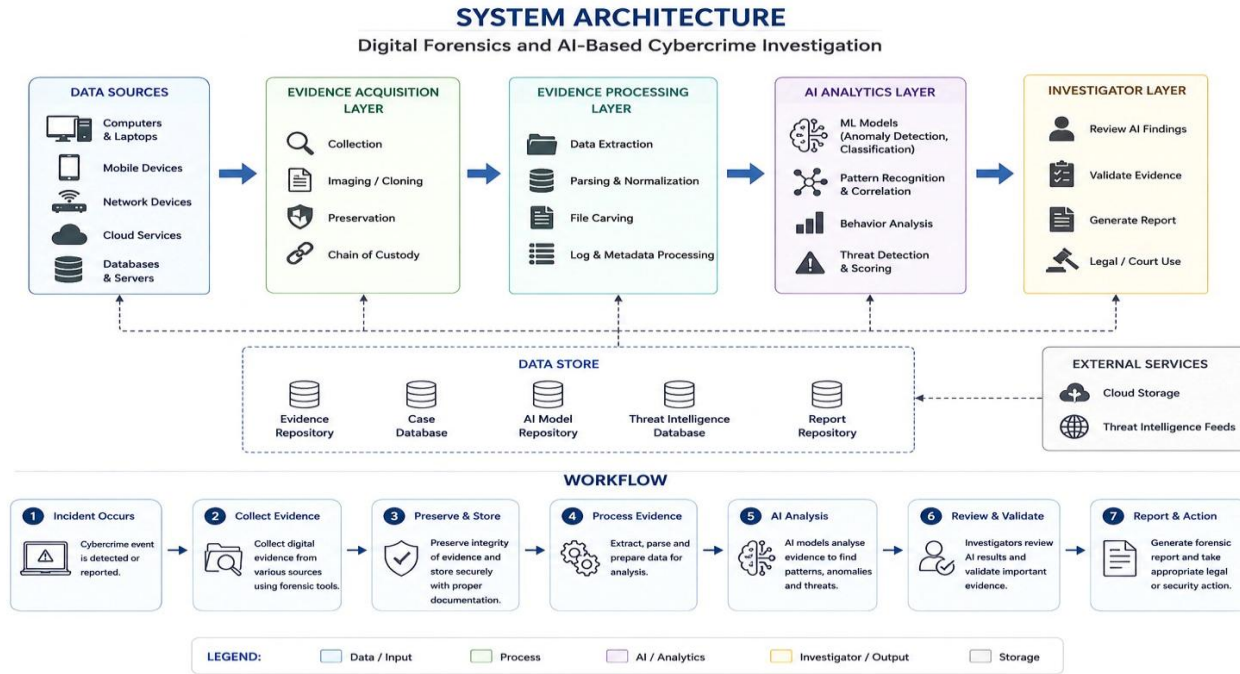
- Digital forensics is the process of collecting and analysing digital evidence from electronic devices.
- It helps investigators identify how a cybercrime occurred and who may be responsible.
- Artificial Intelligence can automate the analysis of large volumes of digital evidence.
- AI-based investigation improves the speed and accuracy of cybercrime detection.

Need and Motivation

- Increasing cyberattacks generate huge amounts of digital evidence.
- Manual investigation of large datasets is time-consuming and complex.
- AI can identify suspicious patterns and connections faster than traditional methods.
- Faster investigations help organisations respond effectively to cybercrime.



Working Principle / Architecture



• Key Components and Technologies •

- Digital Forensic Tools: Used for evidence acquisition, recovery and analysis.
- Artificial Intelligence & Machine Learning: Used for pattern recognition and anomaly detection.
- Data Analytics: Helps analyse large volumes of logs and digital evidence.
- Cybersecurity Technologies: Network monitoring, malware analysis and threat intelligence support investigations.



• Applications / Real-World Use Cases •

- Investigation of malware, ransomware and phishing attacks.
- Recovery and analysis of deleted or hidden digital evidence.
- Detection of suspicious network activity and unauthorised access.
- Investigation of financial fraud, identity theft and online crimes.



Advantages and Limitations

Advantages:

- Faster analysis of large volumes of digital evidence.
- Improved detection of hidden patterns and suspicious activities.
- Automation reduces repetitive manual investigation tasks.
- Helps investigators prioritise important evidence.

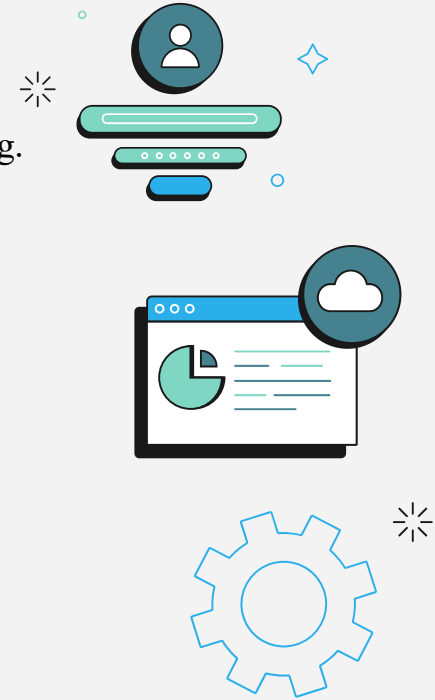
Limitations:

- AI may produce false positives or incorrect results.
- High-quality training data is required for reliable AI models.
- Encrypted or damaged evidence can be difficult to analyse.
- AI-generated findings still require verification by forensic experts.

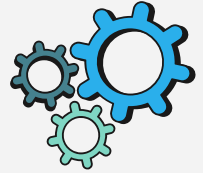


Recent Developments and Trends

- Generative AI is being explored for automated forensic analysis and reporting.
- AI-powered malware and anomaly detection is becoming more advanced.
- Cloud and mobile forensics are receiving greater attention.
- Automated threat intelligence and evidence correlation are improving investigations.



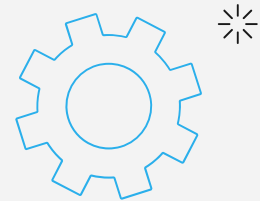
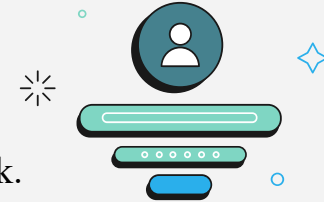
Challenges and Future Scope



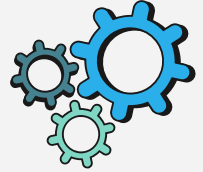
- Maintaining the integrity and admissibility of digital evidence.
- Handling privacy concerns while analysing personal data.
- Detecting increasingly sophisticated AI-generated and automated cyberattacks.
- Future systems can provide real-time AI-assisted forensic investigation.

Case Study / Example

- A ransomware attack compromises an organisation's computer network.
- Forensic investigators collect system logs, files, network traffic and malware samples.
- AI analyses the evidence to identify unusual activities and possible attack patterns.
- Investigators use the results to determine the attack timeline and source of compromise.



Conclusion and References



Conclusion:

- Digital forensics plays an important role in modern cybercrime investigation.
- AI improves the speed and efficiency of analysing digital evidence.
- Combining forensic expertise with AI provides stronger cybercrime investigation capabilities.
- Human investigators remain essential for validating evidence and making final decisions.

References:

- NIST – Guide to Integrating Forensic Techniques into Incident Response (SP 800-86)
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- INTERPOL – Digital Forensics
<https://www.interpol.int/How-we-work/Innovation/Digital-forensics>
- NIST – Artificial Intelligence Risk Management Framework (AI RMF 1.0)
<https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10>
- NIST – AI Resource Center (AIRC)
<https://airc.nist.gov/>